Course syllabus for

Academic year 2022/2023 >

TIF101 - Applied quantum physics

Tillämpad kvantfysik

Course syllabus adopted 2021-02-26 by Head of Programme (or corresponding)
Owner: TKTFY
4,5 Credits
Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail
Education cycle: First-cycle
Main field of study: Engineering Physics
Department: 16 - PHYSICS

Teaching language: Swedish
Application code: 57124
Open for exchange students: No
Block schedule: B+
Maximum participants: 120
Only students with the course round in the programme overview

Credit distribution

Module				Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates		
0119	Examination	4,5 c	Grading: TH		4,5 c					Wed 11/01-2023 am J,	Tue 04/04-2023 am J,	Wed 23/08-2023 pm J

In programs

TKTFY ENGINEERING PHYSICS, Year 3 (compulsory)

Examiner:

Anders Hellman

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Eligibility

General entry requirements for bachelor's level (first cycle)
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Prerequisites are basic quantum physics knowledge, gained for example through the course Quantum Physics (FUF040), and knowledge of mathematics and mechanics corresponding to the basic courses in mathematics and mechanics.

Aim

This course is a continuation of the first course in Quantum Physics (FUF040). Here quantum physics is again applied on elementary phenomena, its rôle for the technical physics is further elucidated including modern technological applications. Its realization in a number of different important systems is noticed through exercises and projects.

Learning outcomes *(after completion of the course the student should be able to)*

- compute energy levels for systems
where approximation methods must be used. - solve time-dependent systems.
- describe the line of thought between the fundamental
Schrödinger equation and the rule to build up the periodic system of the
elements. - describe chemical bonding from quantum mechanical principles. - have gained some experience in choosing suitable model and
approximation for treatment of specific quantum systems.

Content

This course is a continuation of the the first course in Quantum Physics (FUF040).
- Here quantum physics is applied on elementary phenomena
- Interactions of degrees of freedom in many-electron atoms are discussed (spin-orbit interaction, fine structure, Pauli principle etc)
- The periodic table of the elements and the electronic structure of the elements are discussed
- Approximation methods for many-electron systems and time-dependent perturbation theory is introduced and applied.
- The physics of molecules is introduced (types of molecular binding, molecular orbitals, diatomic molecules, molecular spectra, dynamics).
- The role of quantum physics for the technical physics is further elucidated including applications.

Organisation

Lectures and problem solving classes are given. The course has one compulsory experimental excersice.

Literature

"Introduction to Quantum Mechanics" by David J. Griffiths

Examination including compulsory elements

Home assignments, experimental work and written exam.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.